Castles in the Air

Extended chapter commentary

CHAPTER 7
Seekers and serpents

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CHAPTER 7 COMMENTARY

Seekers and serpents

As we explained in Chapter 3, one of our main aims in this book is to see what happens when we use mathematics to study itself. Can the basic ingredients of mathematics — such as mathematical statements, theorems, proofs and computations — themselves be defined and studied in a mathematically precise way? And what can we learn about these concepts by doing this?

We have made a great deal of progress in the last three chapters. So now is a good time to take stock of what we've learned, and to draw up a checklist of our 'hits and misses' so far.

Can we make precise the idea of a 'mathematical statement'?

✓ The *glyph language*, which we introduced in Chapter 3, provides a compact and powerful system for expressing mathematical statements which is itself mathematically precise. Strictly speaking, many different dialects glyph language, with different 'vocabularies', may be needed for talking about different areas of mathematics (the one we've introduced can be used to talk about either Whole numbers or Real numbers). But the same basic grammatical machinery, consisting of just six logical operations, suffices in practice for all of these languages.

A question we haven't addressed yet is whether we really need all these dialects, or whether there's some *single* language that can subsume all of these. We'll return to this in Chapter 9.

Can we make precise the idea of a mathematical problem that can be solved by an 'automatic method' or 'mechanical recipe'?

✓ Our *tube machine* kit from Chapter 4 gives a way of building machines for performing mechanical calculations with Whole numbers. Since many other things (like glyph statements, or proof trees, or even tube machines themselves) can be *coded* as Whole numbers, tube machines in practice give us a way of calculating with these things as well. *Church's Thesis* is the claim that any 'mechanical recipe' for calculating with Whole numbers can be

simulated by a tube machine. The case for Church's Thesis (in the version that concerns us here) is very strong, and no one today seriously doubts it.

Is there a single machine that's able to solve all 'mechanically solvable' problems?

✓ Yes, this is what Turing's *universal machine* does (Chapter 4). The machine needs to be 'programmed' with an input to tell it which problem it's supposed to solve, and this is the key idea at the heart of modern computers.

Can every precisely defined mathematical problem be solved by a mechanical method?

× No. Turing showed that the *halting problem* — the problem of deciding whether a given computation ever finishes — cannot itself be solved mechanically (Chapter 4). We can use this to prove that several other problems are mechanically unsolvable, by showing that if we *could* solve them, we'd have a way to solve the halting problem too.

Is there a purely mechanical method for working out whether a given mathematical statement is true?

✓ For glyph statements about *Real* numbers, yes. This is what Tarski's Real Number Decider does (Chapter 3). The same idea can be extended fairly easily to statements about Complex numbers or about Euclidean geometry.

× For glyph statements about *Whole* numbers, no. As we saw in Chapter 4, there is no *Whole Number Decider*. If there were, we could apply it to glyph statements about whether computations ever finish or not, and so solve the halting problem. The fact that Whole number mathematics is 'harder' than Real number mathematics in this sense is one of the early surprises that the subject has to offer.

Can we make precise the 'laws of logic' that govern mathematical statements?

✓ As we saw in Chapter 5, the *Rules of Thought* for each of our six logical operations provide precise and mechanizable rules by which glyph statements can be logically 'deduced' from other glyph statements. Furthermore, Gödel's *Completeness Theorem* reassures us that our rules form a complete set: there can be no 'purely logical' deductions that can't be carried out with these rules.

Can we make precise the idea of a 'mathematically provable' statement? And is every statement either provable or disprovable?

This is where we've got to just now. For the time being, we're concentrating on statements of Arithmetic (Whole number glyph statements), because even here there's plenty to occupy us. We've seen that our Rules of Thought give us a precise mathematical concept of a *proof* starting from certain basic assumptions — but what 'basic assumptions' about Whole numbers should we choose to start from? Can we find some reasonable set of assumptions that provide a sufficient basis for proving *any* statement of Arithmetic that's provable at all? These are the questions we are investigating in this chapter, taking the system of *Peano Arithmetic* as a convenient starting point.

Of course, the two 'misses' recorded in the above list shouldn't be seen as failures. In fact, they're among the most striking successes that we've achieved so far, because in both cases, we've not merely failed to do something — we've actually proved that it *can't* be done. So overall, we've been getting on remarkably well.

Actually, I was really using the ticks and crosses to highlight another point. We can think of the discoveries marked with ticks as supporting the general idea that the big scary beast of mathematics is *tameable*: that is, they show how certain seemingly rich and complex notions may be analysed and explained in terms of some small set of basic elements — be they logical symbols, tube components, or Rules of Thought. This is just the sort of thing that mathematicians (and scientists in general) like to do: to explain some large and complex class of phenomena using just a handful of basic principles. By contrast, the discoveries marked with crosses indicate that the beast also has its wild side: that some aspects of its nature *can't* be tamed in the way one might have hoped. This tension between the broadly opposing forces of 'tameness' and 'wildness' is big part of what makes mathematical logic the subject it is. So far, to judge from the list above, the balance may seem to be tilting towards the tame side — but we haven't finished yet!

Let's now turn to the ideas we met in this chapter's story episode. After an important philosophical prelude, we'll look more closely at the system known as Peano Arithmetic, then move on to the subtle and slippery world of Gödel's incompleteness theorems and their ramifications.

The surveyability principle

We've asked what basic axioms about Whole numbers we should choose as the starting point for our proofs, and I began our story by introducing one possible set of axioms — those of *Peano Arithmetic*, or PA for short — which has a great deal going for it. The system is so named after the Italian logician Giuseppe Peano who presented a version of it in 1889, building on profound

ideas of Richard Dedekind.[1] (Notice that 'Arithmetic' by itself refers just to the *language* of Whole number glyph formulae, whereas 'Peano Arithmetic' refers to a system of axioms formulated in this language. Incidentally, 'Peano' is pronounced as three syllables, a bit like 'piano'.)

Before we get into the details of PA, I should explain that for most of this chapter we'll be proceeding from a less 'sceptical' position than the austere Hilbertian one which we discussed in the last chapter. Whereas for Hilbert, even the statement 'there are infinitely many primes' (taken by itself) was meaningless, we shall here take it for granted that this, and indeed any statement we can express in Arithmetic, has a clear meaning, so that it makes sense to ask whether such a statement is true or false.

What is the philosophical stance that underpins this more 'generous' view? What differentiates it from Hilbert's position? Can we identify some fundamental philosophical principle which, if we adopt it, will imply that all Arithmetical statements are meaningful?

Indeed we can. In fact, this will form one of the central philosophical ideas of this book, so it's worth taking a bit of time over it. I'll first state what the principle is, then explain the intuition behind it, and finally outline how it implies that all Arithmetical statements are meaningful. I should make clear that I'm using 'meaningful' here in the specific sense of 'having a definite, objective truth value' — that is, being objectively either 'true' or 'false'.

Here, then, is a statement of the philosophical principle, or 'belief', which I invite you to consider.

SURVEYABILITY PRINCIPLE FOR WHOLE NUMBERS:
If P *is any formula with the single free variable* x*,*
and if for every individual numeral n*, the statement* P[x→n] *is meaningful,*
then the statement ∃x P *is also meaningful.*

For example, suppose P is a glyph formula that says 'x is a counterexample to the Goldbach conjecture' (that is, x is an even number greater than 2 that's not a sum of two primes). Since this is a Checkable property, we'll take it as uncontroversial that for any *particular* value of x, P makes a meaningful statement (which may be either true or false). The surveyability principle then invites us to conclude from this that the statement 'there exists x such that P' (i.e. there exists a counterexample to the Goldbach conjecture) is itself either true or false. Of course, it doesn't tell us *which* it is — we still don't know the answer to this! — but it claims that there is some objective 'fact of the matter' about whether the Goldbach conjecture is true. Intuitively, it's saying something about what we'd be able to observe if we could 'see all the Whole numbers at once'. We'll often refer to this principle more briefly as the *surveyability of* ℕ, where ℕ is the usual mathematical symbol for the set of all Whole numbers (also known as *natural numbers*).

We can present the idea even more simply than this. Imagine a row of tiny light-bulbs numbered 0, 1, 2, …, starting just in front of you and stretching off to infinity. Each bulb is definitely either on or off. We can think of the surveyability principle simply as saying that there's a definite 'fact of the matter' about whether at least one of the bulbs is on. That is, *either* some bulb is on *or* all of them are off — regardless of whether we have any way of *knowing* which of these is the case.

You might feel this idea is so 'obvious' that you find it hard to see how anyone could think otherwise. So let's see if we can get our minds inside the alternative 'sceptical' view. The argument is one we hinted at in the last chapter: it claims that the surveyability principle is based on an unwarranted extrapolation from what is true for just a *finite* number of bulbs. After all, what do we *mean* by the question 'Are any of the lights on?' One view would be that (in the finite case at least) we know what we'd have to do to answer this question — just check each of the bulbs in turn — and that this verification process *constitutes* the meaning of the question: what the question is really asking is 'What would the result of such a process be?' So the meaning of the question would have to amount to 'If we could check all the bulbs in turn, what would we find?' But of course, in the infinite case, we can't do this. And it's on precisely this ground that the sceptic would reject the question as meaningless.

One might try to counter this with the idea that it's at least *logically* possible that God, or an angel, could check the infinitely many bulbs and report the answer to us. One can have some fun here with thought experiments in which the bulbs 0, 1, 2, 3, … are respectively checked at times 0, 1/2, 3/4, 7/8, …, so that God knows the answer by time 1. Expressed in these terms, the essence of the surveyability presence is that *it makes sense to talk about what the answer 'would' be* if we did this. But the sceptic will be unimpressed by all this fantasizing. It is simply a cognitive illusion (the sceptic might say) that in the infinite case there is necessarily some 'fact of the matter' about whether one of the bulbs is on. Unless we have some *human* way of knowing this, we have no right to say that any such fact of the matter 'exists'.

This point of view was forcefully argued by the Dutch mathematician L.E.J. Brouwer, whose distinctive philosophy we shall look at in Chapter 11. But a particularly eloquent expression of the problem comes from the philosopher Ludwig Wittgenstein, drawing on Brouwer's ideas:

> "In the decimal expansion of π either the group '7777' occurs, or it does not — there is no third possibility." That is to say: "God sees — but we don't know." But what does that mean? — We use a picture; the picture of a visible series which one person sees the whole of and another not … And this picture seems to determine what we have to do, what to look for, and how, but it does not do so … Here saying "There is no

> third possibility" … expresses our inability to turn our eyes away from this picture.[2]

There *are* indeed mathematicians and philosophers who don't accept the surveyability of $\mathbb{N}$. To many of these, even my efforts to explain what the surveyability principle *is* would probably appear as meaningless or nonsensical. In an effort to get the idea across, we might reach for some slogan such as 'God can see all the Whole numbers at once'. We might, of course, qualify this by saying this was only a manner of speaking rather than a literal statement of theological belief — but even so, for the sceptic, this use of language might only serve to reinforce the impression that the principle is somehow 'metaphysical' in character. Indeed, insofar as it asserts the existence of some truth independently of our ability to know it, the surveyability of $\mathbb{N}$ can be seen as a 'platonistic' belief, albeit perhaps a relatively mild one. In the light of this, we can perhaps see why Hilbert, working within a philosophical and cultural climate of extreme scepticism and suspicion, was so eager to demonstrate that no such 'article of faith' was needed as a starting point for mathematics, but that mathematical reasoning could be grounded in more primitive, concrete and down-to-earth concepts.

Let's now see how the surveyability principle leads to the idea that all Arithmetic statements are meaningful. We've seen that all Arithmetic formulae are built up from *basic* formulae (involving $=$, $<$ or $\leq$) by means of the six logical operations. So broadly speaking, the idea is this. Any basic statement of Arithmetic will be meaningful because it is Checkable: whether it's true or false can be determined by direct calculation. And if P and Q are any meaningful statements, we can also say that $P \,\&\, Q$, $P \vee Q$, $P \Rightarrow Q$ and $\neg P$ are meaningful, because their truth-values will be determined from those of P and Q using the truth tables from Chapter 3. So that just leaves $\exists$ and $\forall$. For $\exists$, the surveyability principle gives exactly what we need: as long as all individual instances of P are meaningful, we can conclude that $\exists x\, P$ is meaningful. And the same will apply to $\forall$, because it's possible to rephrase $\forall x\, P$ as $\neg\exists x \neg P$. We can therefore show that *all* Arithmetic statements are meaningful by starting with basic statements and working our way up to successively larger ones using these observations.

I'm not especially trying to argue *for* the surveyability of $\mathbb{N}$ at this point, but simply to point out the philosophical premise whose consequences we shall be investigating over the next few chapters. Since both Anna and Emily currently accept without question that all Arithmetic statements are meaningful, we may suppose that they implicitly accept the surveyability of $\mathbb{N}$, and are exploring how the world looks on that basis. Later, in Chapter 11, we shall again call this belief into question and see where that leads. And even

within the present chapter, we shall make the connection with Hilbert's programme and conclude the story we began in Chapter 5.

With these philosophical ideas in place, let's now review the various axioms of PA, grouping them slightly differently from how they appeared in the story. The idea is that if we accept the surveyability of $\mathbb{N}$, each of these axioms says something that we'll readily recognize as true. So we can think of PA as making a good first stab at what we'll be led to believe about Whole numbers if we accept this principle.

The axioms of Peano arithmetic

We'll begin with some axioms that express basic facts about '='. These axioms aren't really specific to Whole numbers as such: they would be just as applicable to Real numbers or to anything else for which there's a sensible concept of 'being equal', and for this reason, they're sometimes treated as part of pure logic rather than mathematics. First, we have the so-called *reflexivity* axiom, which just says that every Whole number (or whatever) is equal to itself:

$$\odot \quad \forall x \quad x = x$$

Second, there's the infinite family of *substitution axioms* generated by the second minting machine, which tell us that if x = y then we're allowed to replace 'x' by 'y' in any formula we like. We saw a specific example of such an axiom in the story, but let's now look at the general form for such axioms.

To create such an axiom, we need to supply a formula P, which contains a special variable z marking the points at which we're wanting to replace 'x' by 'y'. Using the notation we met in the previous chapter, we can then write P[z→x] for 'P with z replaced by x', and P[z→y] for 'P with z replaced by y'. For readability, we'll abbreviate these to P[x] and P[y] respectively. So we can write the substitution axiom for P as:

$$\odot \quad \forall x \, \forall y \quad (x = y \;\; \& \;\; P[x]) \;\Rightarrow\; P[y]$$

Notice that what's written above isn't *itself* an axiom of our system in the way that the reflexivity axiom is. This is because the symbol P isn't itself part of our language of Arithmetic: rather, as we saw in Chapter 5, it's a metavariable that can *stand for* any glyph formula we please. So we should think of the expression above not as a specific axiom but as an *axiom scheme*: that is, a sort of template from which it's possible to run off an unlimited number of particular axioms, one for each choice of P. That is exactly the purpose of the minting machine in the story.

It turns out that reflexivity and substitution give all the axioms for '=' that we ever need. For instance, using these axioms along with the Rules of Thought, it's possible to prove two important facts about '=', known respectively as *symmetry* and *transitivity*:

⊙ $\forall x\ \forall y\ \ x = y \Rightarrow y = x$ ⊙ $\forall x\ \forall y\ \forall z\ \ (x = y\ \&\ y = z) \Rightarrow x = z$

You might enjoy seeing whether you can construct proof trees for these. They are not large, but perhaps not trivial to discover. Solutions are given in the supplement on *Formal proofs in pure logic* on the book's website.

Now for the axioms that try to get to the heart of what Whole numbers actually are. The basic idea we're trying to capture here is that all the Whole numbers can be *generated* by starting from 0 and repeatedly applying the operation of adding 1. First, we have two axioms giving important properties of this '+1' operation:

⊙ $\forall x\ \ \neg\, x + 1 = 0$ ⊙ $\forall x\ \forall y\ \ x + 1 = y + 1 \Rightarrow x = y$

In effect, these say that the '+1' operation always produces a number different from 0 itself, and that if applied to two *different* numbers x and y, this operation always produces two different numbers x + 1 and y + 1. (This slightly back-to-front rephrasing of the second axiom might help to convey the intention behind it.)

Besides these, we'd also like to capture the idea that there are no Whole numbers *other* than those generated by 0 and +1: that is, that every Whole number can be 'reached' by means of these two operations. This is the idea behind the *induction axioms* that are generated by the third minting machine, which call for a little more explanation.

The essence of mathematical induction isn't hard to grasp. The basic idea is that one way to show that something is true for *all* Whole numbers is to show, first, that it's true for the number 0, and second, that whenever it's true for a number x, it's also true for x + 1. By combining these two facts, it's easy to see that the property in question must hold for the number 1, from which it follows that it must also hold for 2, and hence for 3, and so on. Indeed, because all Whole numbers can be reached from 0 by applying +1 some number of times, these two facts will be enough to guarantee that the property holds for all of them. This fundamental principle is used ubiquitously in number theory and in very many other areas of mathematics.[3]

Two further metaphors might be helpful here. The first (which has popped up occasionally in our story) is that of climbing an infinite ladder: if we know how to get onto the bottom step, and we know how to get from any step to the one above it, then we're able to climb as high as we like. The other is to think, rather loosely, of numbers as being 'infected' by some contagious property

that can be passed from one number to another. If 0 is infected, and the property will be passed from any infected number x to x+1, then all numbers will become infected.

This simple principle is what is captured by the induction axioms. We can conveniently express the idea via an axiom scheme, using the metavariable P for the property we're interested in. As before, we will use a special variable z as a place-marker for the Whole number to which the property refers, and will abbreviate P[z→x] to P[x], P[z→0] to P[0] and so on.

$$\odot\ P[0]\ \&\ (\forall x\ P[x] \Rightarrow P[x+1])\ \Rightarrow\ (\forall z\ P[z])$$

Once again, this should be seen as a template from which an unlimited number of axioms can be run off, one for each property P. For example, in the story Anna applied the axiom minter to the rather trivial property '$3 < z + 4$', which resulted in the axiom:

$$\odot\ (3 < 0 + 4\ \&\ (\forall x\ 3 < x + 4 \Rightarrow 3 < (x + 1) + 4))\ \Rightarrow\ (\forall z\ 3 < z + 4)$$

The induction axioms, together with the preceding two axioms for +1, form the heart of our system, since these are the axioms that try to capture the core of our understanding of what Whole numbers 'are'. It's these axioms in particular that are usually referred to as the 'Peano axioms'.

I say 'try to capture' quite purposefully here, because there's a rather subtle point that's worth flagging up. We've already mentioned that we're trying to capture the idea that all the Whole numbers can be generated from 0 and +1 — that is, there are no other Whole numbers besides the ones these give rise to. Let's call this the *generation principle* for Whole numbers. However, there is a problem: there's actually no good way to state this generation principle *directly* within our language of Arithmetic as it currently stands. For the time being, then, the best we can do is to frame our induction scheme, which, as we've seen, expresses an important *consequence* of this generation principle. This rather subtle distinction will come back to haunt us in Chapter 10.

Having done our best to capture the essence of Whole numbers, all that remains is to add a few further axioms which, in effect, serve as *definitions* for the remaining symbols of the language. For instance, we've so far treated '+1' as a basic operation, but to pin down the meaning of '+' in general, we introduce the following axioms:[4]

$$\odot\ \forall x\ \ x + 0 = x \qquad\qquad \odot\ \forall x\ \forall y\ \ x + (y + 1) = (x + y) + 1$$

The idea here is that these determine the meaning of '$x + y$' for all Whole numbers x and y, because all possible numbers y are generated via 0 and +1. As we saw in the story, there are also some similar axioms that pin down the

meaning of '×', and a few more that serve as definitions of '<' and '≤'. Finally, we want definitions for the numerals 1, 2, 3, 4, 5, …, and this purpose is served by the first minting machine which produces axioms such as

$$1 = 0 + 1\,, \quad 2 = 1 + 1\,, \quad 3 = 2 + 1\,, \quad 4 = 3 + 1\,, \quad 5 = 4 + 1\,, \quad \ldots$$

Why axioms — and why these?

Hopefully, our survey of the axioms of PA will have convinced you that they all say things that are unquestionably true of Whole numbers and the usual operations on them, assuming we accept that these axioms are meaningful at all (which follows from the surveyability of $\mathbb{N}$). Nonetheless, there are several questions one might naturally ask at this point.

Firstly, you might wonder whether we really need axioms at all, in the sense of unproved assumptions about Whole numbers. Wouldn't it be more satisfactory if we could somehow *define* Whole numbers in some way that didn't involve assumptions, then prove that such numbers do indeed have the properties that we're here calling 'axioms'? After all, isn't mathematics all about proving things — about achieving complete certainty? If so, isn't it something of a blot on our system that there are axioms we just have to 'take on trust' at the very outset?

There's an interesting story to be told here: namely, that of the attempts by Bertrand Russell and others to ground the axioms of PA (and similar systems) in pure logic alone. We'll come back to this in Chapter 12; but for the time being, suffice it to say that we shall make faster headway if we simply accept PA as our starting point.

A second, more profound, question is whether the axioms of PA are *all* the axioms about Whole numbers that we'll ever need. In the previous chapter, we not only introduced a system of Rules of Thought, but saw that there was a theorem saying that these rules formed 'a complete set'. Is something similar in the case of PA? Or are there other unquestionably true properties of Whole numbers which PA doesn't capture?

One of the main ideas we're looking at in this chapter is that PA is *incomplete,* in the sense that there are statements that PA can neither prove nor disprove. But that doesn't in itself settle the question of whether PA is 'the best we can do', or whether there's some other set of unquestionably true axioms that can prove more than PA. At this point in the story, our heroines are divided in their opinions on this, but in the next chapter we shall discover a definitive answer to this question.

Thirdly, one might ask: What if we *don't* accept the surveyability of $\mathbb{N}$? What if we're specifically wanting some formal system that captures only the

modes of reasoning accepted by the Hilbert school? Can we achieve this via some suitably cut-down version of PA?

The short answer is that we can. I won't give all the details here, but it's worth indicating the general idea, as we'll be wanting to refer to such a system later in the chapter.

There has been some interesting debate over exactly where the boundary of 'finitistic reasoning' should be drawn, but the mainstream view nowadays is that Hilbert's view (as expressed in his papers from the 1920s) is adequately represented by a system known as *Primitive Recursive Arithmetic* (PRA).[5] To cut down from PA to PRA, we need to impose a radical restriction to ensure that we're working entirely with Checkable or Predictive statements — the only kinds that are meaningful from Hilbert's perspective. One clean way to do this is via the drastic requirement that *all formulae used*, whether in axioms, theorems or proofs, *must be Checkable*. For instance, we can use the Checkable formula $x + y = y + x$ to do duty for the Predictive statement $\forall x \forall y \; x + y = y + x$, not bothering with the $\forall$ prefixes. We can get away with this simplification because, after all, to have *proved* $\odot \; x + y = y + x$ means to have shown that $x + y = y + x$ regardless of what x and y are.

Once we've adopted this draconian measure, we can adopt all our usual Rules of Thought to the extent that they apply to Checkable formulae. Beyond this, there are a few minor points to attend to in order to obtain a system of the required strength, but we won't need to go into these here.[6] The main point is simply that this can all be done, and that it results in a system for finitist mathematics which we'll call PRA. It's also not hard to check that everything provable in PRA is provable in PA.

We shall come back to PRA towards the end of the chapter when we resume the story of Hilbert's programme. But our main focus for the time being will be on PA. Having described the nuts-and-bolts of this system, let's see what it has to offer and what can be proved in it.

The power of PA

PA gives excellent value for money. As Anna and Emily found, its simple set of axioms gives all the information about Whole numbers we need in order to construct rigorously formal proofs for a vast body of interesting theorems: Euclid's theorem on the primes, Fermat's little theorem, and much, much more. Even in cases such as Dirichlet's theorem, where the usual proof makes use of concepts outside the scope of Arithmetic, it's often possible to massage the proof into a form that fits within PA — although admittedly at the expense of clarity and intelligibility.

Likewise, many other theorems whose very *statements* might not seem to fit within Arithmetic, because they talk about things other than Whole

numbers, can often be replaced by equivalent statements that can, using various 'coding' tricks like those we met in Chapter 4. In this way, PA turns out to be powerful enough for a great many theorems in areas such as finite graph theory, finite combinatorics, and indeed mathematical logic itself.

An interesting test case is provided by Wiles's monumental proof of Fermat's Last Theorem. As we saw in Chapter 2, this takes us through all kinds of weird and wonderful mathematical territory, far exceeding Dirichlet's proof. Can even this stupendous argument be in principle recast in nothing but PA? At present, the consensus of those best placed to judge is that it almost certainly can, although a lot of detailed and difficult work would be required in order to confirm this definitively.[7]

There are certainly major areas of mathematics that PA *doesn't* naturally cover: these would include most of Real and Complex analysis, or indeed any kind of mathematics that makes serious reference to *infinite* objects or structures. (As Emily noted, even a single Real number can in general involve an infinite amount of information, which no single Whole number ever can.) But the problem, in these cases, is merely that the *statements* in question exceed what we can say in Arithmetic. (As we indicated in Chapter 3, there may be other dialects of glyph language that are suitable for them.) In other words, the inability of PA to deal with these areas just reflects the limitations of the language, rather than of the actual means of proof. If you can get as far as *stating* your theorem in Arithmetic, and if you can prove it by any means at all, then there's an excellent chance that you'll be able to prove it in PA.

Indeed, one might well be tempted to make the following conjecture, which I shall call *Emily's thesis*:

Any theorem which can be stated in Arithmetic,
and which can be proved at all, can be proved in PA.

If one were to look at the vast body of theorems discovered to date by mathematicians pursuing the traditional goals of number theory (as opposed to those of mathematical logic), the cumulative evidence for this conjecture would seem very strong indeed. Although, as we saw in Chapter 2, there have been theorems that were first proved via long 'detours' into Complex number land or other territory, the passage of time has often shown how such detours can be avoided — and even in cases where this isn't yet clear (as in the proof of Fermat's last theorem), it is generally expected to be possible.[8] So it would seem that PA makes a good stab at capturing the notion of a 'mathematically provable theorem', at least as far as Whole number statements are concerned.

Of course, Emily's thesis would be automatically true if *every* true statement of Arithmetic were provable in PA. Unfortunately, however, the latter is not the case — and this brings us to the main business of the chapter:

the two *incompleteness theorems* which Kurt Gödel established in his epoch-making paper of 1931.[9]

The Gödel effect

Gödel's incompleteness theorems are remarkable. The easy things to say are that they're among the greatest mathematical discoveries of the 20th century, that they're the most momentous results in the whole of mathematical logic to date, and that they lie at the absolute heart of the story we're telling in this book. But there's something else that's harder to pin down. Gödel's theorems capture the imagination. They produce powerful resonances within the psyche. And some people find them deeply unsettling.

I can remember first reading Gödel's paper when I was 17, in the course of an overnight train journey with my brother from London to Inverness. I had already read a few popular accounts of Gödel's ideas, but it was only when I worked through his own presentation that I properly understood them. At the time, it seemed to me as though some deep core belief had been shaken — almost as though the whole basis of human rationality had been undermined. I felt physically nauseous, and even the buildings outside the window seemed to lean threateningly as though lacking a secure foundation.

So I want to begin by reassuring you that it's really not *that* bad! Gödel's theorem's might indeed seem disturbing and perplexing on a first encounter; but once you've digested them and lived with them for a bit, they do make perfect rational sense. There's nothing genuinely 'paradoxical' about them, and they certainly don't come anywhere close to undermining 'rationality itself'. Having said that, they may have a tendency to stimulate thoughts that do run in that direction; so if you want my advice, an overnight train journey probably isn't the best time for learning about them.

The peculiar allure of Gödel's theorems is reflected by the sense of mystique that frequently surrounds them. Few other theorems in mathematics, and certainly no other theorems in mathematical logic, have ever appealed to a wide public in quite the same way. But because a genuine understanding of these theorems isn't so easy to acquire, the effect of this is that all kinds of garbled or half-true versions of these theorems have entered the popular consciousness, with some quite extraordinary results. Gödel's theorems have found their way into science fiction, music, photography, feminist theory — and more recently, into Internet discussion groups on just about every topic under the sun. As the logician Torkel Franzén noted:

> In any such group … somebody will sooner or later invoke Gödel's incompleteness theorem. One finds such invocations not only in discussion groups dedicated to logic, mathematics, computing, or philosophy, where one might expect them, but also in groups devoted to

> politics, religion, atheism, poetry, evolution, hip-hop, dating, and what have you.

Why should this be? What is it about these theorems that fascinates people? I suspect there are two factors. Firstly, it would seem that something in the human mind is deeply attracted by the concept of self-reference or self-application in general: the idea of 'the serpent biting its own tail'. The all-time supreme eulogy to this idea is surely Douglas Hofstadter's *Gödel, Escher, Bach: An Eternal Golden Braid* — a book which made a huge impression on me as a teenager. But secondly, the theorems themselves are richly suggestive, apparently hinting at some deep principle that *surely* ought to have significance beyond the purely mathematical: that there are inherent limits to what we can know, or something of the kind. What's more, being mathematical theorems, they carry the authority of something one can't argue with. So the temptation seems almost irresistible to enlist Gödel's theorems in support of all manner of arguments in philosophy, psychology, literary theory, politics and elsewhere. The whole phenomenon is explored in detail in Franzén's book *Gödel's Theorem: An incomplete guide to its use and abuse*.[10]

Not all of this is bad. I'm certainly not saying that one shouldn't look for applications of Gödelian ideas outside mathematics, or even for loose analogies elsewhere. I do think that the incompleteness theorems are of great philosophical importance, and something would be lost if we only ever approached them in what the logician Jean-Yves Girard called 'a narrow-minded technical spirit'. But if one wants to invoke Gödel for a reasoned argument in support of anything, a thorough understanding of the ins-and-outs of the theorems is absolutely essential. For this reason, much of what follows will be devoted to highlighting some of the major pitfalls and potential sources of confusion, and in particular (at the risk of labouring the issue) to pointing out a number of things the theorems *don't* say. I'm also happy to recommend two excellent and careful books that undertake these tasks more thoroughly than I've done here: the book by Torkel Franzén already mentioned, or (for more technical detail) Peter Smith's *An introduction to Gödel's theorems*.

The first incompleteness theorem

Here's how we stated the first incompleteness theorem in our story:

> *Let* **T** *be Peano Arithmetic, or*
> *any extension of it with a computably defined set of axioms.*
> *If* **T** *is consistent, then* **T** *is incomplete.*

There are several points here to be unpacked, but let's begin with the very last word. We say a formal system such as PA is *incomplete* if there are glyph statements that can be neither proved nor disproved in the system. Such statements are sometimes referred to as being *undecidable* in the system; however, there's an unfortunate source of confusion here, because the word 'undecidable' also has a well-established usage in another sense, as we saw in Chapter 4. In this book, to reduce confusion, I'll use the term *unresolvable* for statements that are neither provable nor disprovable in some given system. So the theorem tells us that assuming PA is consistent, there are Arithmetical statements that are unresolvable in PA.

We presented the first incompleteness theorem in the story using PA as an example. However, it's important to understand that the essence of the theorem is by no means limited to PA, but will apply equally well to any formal system of a broadly similar character. Indeed, part of the beauty of Gödel's ideas is how *robust* they are: we can't get round the theorem just by making the right tweaks to our language or rules or axioms. The incompleteness phenomenon is a fundamental and inescapable limitation of *all* attempts to reduce mathematics to a set of symbolic rules, and any or all expressions of this limitation can reasonably be referred to as 'Gödel's theorem'.

Gödel's original proof worked with a particular system known as *Principia Mathematica* (due to Russell and Whitehead), although he pointed out that the same argument would clearly apply to a wide class of other systems.[11] In our statement above, we've presented the theorem as applying to PA and certain extensions of it. However, for the purpose of exploring the ideas, it's worth generalizing our statement just a little further, to avoid any apparent bias towards PA and to clarify what properties of our system are really essential. For convenience, we'll stick for now with systems that use our Whole number glyph language as introduced in Chapter 3 and our Rules of Thought as in Chapter 5. But we'll also informally indicate how even the particular choice of language and rules can be varied without affecting the essential validity of the theorem.

The more general version of the first incompleteness theorem that we shall discuss here is as follows:

Let **T** *be any computably defined set of axioms*
such that every true Checkable statement is provable from **T**.
If **T** *is consistent, then* **T** *is incomplete.*

There are still quite a few bits and pieces here to be explained, so let's take these in turn.

The ‘computability’ condition

Firstly, our statement of the incompleteness theorem applies only to systems of axioms that are *computably defined*. What this means is just that there should be some mechanical way of testing whether a given statement is an axiom or not. (Or a bit more precisely, that there’s a tube machine that can test whether or not a given Whole number is a *code* for an axiom, using the coding of glyph formulae as Whole numbers that we met in Chapter 4.)

This isn’t a very severe restriction in practice, because for any set of axioms that you’re likely to pick — even if it’s some infinite set defined with the help of axiom minters — it will be an easy task to tell whether something is an axiom or not. Indeed, if that were not the case, one could argue that your system wasn’t even doing what formal proof systems are designed to do — namely, to present some class of ‘proofs’ in terms of purely symbolic, mechanically checkable rules. So this ‘computability’ condition would seem to be a fairly innocuous one.

Nonetheless, it’s worth pausing to see what would go wrong if we didn’t include this condition in our theorem. There are two ways to answer this. One way is to point out that the *proof* we described in our story would break down, because this proof relies on being able to build a *Proof Checker* for our system. A Proof Checker, we recall, is a tube machine which checks whether a given number is (a code for) a valid proof tree in our system, and part of the job of checking whether a proof tree is valid will be to check that all the theorems at the leaves of the tree are legal axioms (or else are generated by the Assumption Rule). And this can’t be done unless our set of axioms is computably defined.

But there’s another way of seeing the need for a computability condition which goes rather deeper: our theorem would actually be *false* if we dropped this condition! For instance, here’s a really silly set of axioms that’s not computably defined and which would break the incompleteness theorem: take **T** to be the set of all Arithmetical statements that are, in fact, *true*! (I’m assuming here that we’re happy with the surveyability of $\mathbb{N}$, so that we accept truth for Arithmetic as a well-defined notion.) Of course, this is an utterly useless system in practice, because as we saw in Chapter 4, we have no computable way of telling in general whether a statement *is* true. But that’s exactly why a computability condition is necessary. Indeed, it’s trivial to see that our **T** is a *complete* system of axioms: if P is any glyph statement, then either P is true, in which case it’s present in the set **T** and so certainly provable from **T**, or P is false, in which case ¬P is true and is present in **T**, which means that P itself is disprovable from **T**.

This is an extreme example, but it shows why some condition on the set **T** is needed. Actually, if we’re willing to rework our proof a bit, we can get away with a milder condition than **T** being ‘computably defined’: it’s enough

to assume that **T** is *computably generated*, in the sense that there's some mechanical procedure for listing all the axioms of **T**. We could also extend our result to systems involving other languages and proof rules, provided we also insist that both the axioms and the proof rules were of a 'computable' character. Ultimately, the purpose of the computability condition is simply to ensure that a tube machine playing the role of a Proof Checker can be built, and as long as this is the case, our argument can proceed.

Perhaps even more fundamentally, though, it's important to note that Gödel's theorem speaks only about provability *within formal systems*: that is, in settings where the acceptable methods of 'proof' have been exhaustively specified up front. Once this has been done (in a computable way, and subject to the other conditions of the theorem), it will inevitably turn out that there are statements that slip through the system's net, in the sense that they are neither provable nor disprovable within it. But the incompleteness theorem, as it stands, says nothing about what might be provable if we allow *any means of proof at all* — or more accurately, any means of proof which the human mind can see to confer mathematical certainty. This is because we don't know, at the outset, whether or not our human capacity for proving things can be matched in strength by some suitable formal system, nor what that system might be if it exists. The quest for this particular Holy Grail — the supposed formal system underpinning the human proof engine — will exercise us greatly over the next few chapters.

I emphasize this because there's a natural temptation to simplify the statement of the first incompleteness theorem down to something like 'there are mathematical statements which are neither provable nor disprovable'. This is seriously misleading if it is taken as saying that Gödel's theorem, in and of itself, establishes some fundamental limit to the powers of human mathematical reasoning. The theorem speaks only about the limits of 'computably generated' formal systems. *Whether* the extent of our human mathematical ability can be captured by some such system is an important but separate question.

The 'sufficient strength' condition

Now for the second condition in our statement of the incompleteness theorem: the requirement that *every true Checkable statement* (of Arithmetic) *is provable from* **T**. This condition is again quite a mild one; along with consistency, it serves to ensure that what's provable from **T** doesn't depart too wildly from what we'd ordinarily consider to be true.

Remember from Chapter 5 that *Checkable* statements are a certain class of glyph statements whose truth or falsity can be checked purely by 'routine calculation' — without the need for any real thought, so to speak. So we can

think of true Checkable statements as being in principle trivial to establish, and so it's reasonable to suppose that any axiom system worth its salt should be capable of proving all such statements. Indeed, this much is true not just for PA and larger systems, but even for smaller systems like Hilbert's PRA. Indeed, it's not hard to see intuitively why this should be so: for any true Checkable statement, there's a finite mechanical computation that shows it to be true, and the steps of this computation themselves amount to a 'proof' of the statement which any reasonable logical system will be able to articulate.

Like the computability condition, this 'sufficient strength' condition turns out to be necessary for the proof of the first incompleteness theorem to work, and one can again give examples to show that the theorem would break if the condition were omitted. However, the details are not so important, so I won't dwell on this point.

It's worth pointing out that the condition requires the provability of all true Checkable statements in our standard language of Arithmetic, which includes both '+' and '×'. It's instructive to see what would happen if, for instance, we dropped the '×' operation and the associated axioms. In this case, we'd be left with the system known as *Presburger Arithmetic*, which we met briefly in Chapter 3. One can think of this as a very weak system in which it's not possible to pose any genuinely difficult mathematical questions — too weak, in fact, for the incompleteness phenomenon to kick in. Although Presburger Arithmetic is able to prove all true Checkable statements in the cut-down language without '×', this isn't enough for the proof of incompleteness to go through — and indeed, the usual axioms of PA minus the axioms for '×' do form a *complete* axiom system for Presburger Arithmetic.

The moral, then, is that a system needs to support a certain level of basic arithmetic in order for Gödel's theorem to apply. In the case of systems that use some language other than Arithmetic, what we would typically require is that this language provides some way of *expressing* Checkable Arithmetical statements, and that all true statements of this kind are provable in the system.

The role of consistency

And so to the final ingredient in our statement of Gödel's theorem: the condition that **T** is a *consistent* set of axioms. All this means is that there's no statement P such that both P and ¬ P can be proved from the axioms of **T**.

For our present purposes, an inconsistent set of axioms is as good as useless. Indeed, if you can prove both P and ¬ P from the axioms, then you can go on to prove *any statement you like*, such as 0 = 1. Let's see why this is. Suppose that both ⊙ P and ⊙ ¬ P are provable from **T**. Then it's also possible to prove these statements in the presence the additional assumption ¬ 0 = 1:

$$\neg\, 0 = 1 \odot P \qquad \neg\, 0 = 1 \odot \neg\, P$$

This can be seen either by the Weakening Rule which we mentioned in Chapter 5, or by re-working the proofs of $\odot$ P and $\odot \neg$ P with the extra assumption $\neg$ 0 = 1 slipped in to every line in the proof. In any case, once we've got this far, we can use the Contradiction rule to prove $\odot \neg \neg$ 0 = 1, and then the Double Negation rule to derive $\odot$ 0 = 1.

The same method will clearly work for any statement Q in place of 0 = 1. So if **T** is inconsistent, then knowing that Q is provable from **T** 'proves' nothing at all about Q, and certainly not whether Q is true.

A common mistake when quoting Gödel's theorems is to omit the condition that the formal system must be consistent. But it's very clear why this is necessary: if **T** is inconsistent, then there certainly can't be statements that are neither provable nor disprovable, because every statement is *both* provable *and* disprovable!

The question of consistency is related to the question of *truthfulness*. Let's suppose that we're happy with the philosophical position that every statement of Arithmetic is either *true* or *false*. We can say that **T** is truthful if all the axioms in **T** are, in fact, true. It's easy to see that a truthful system is consistent: if all the axioms of **T** are true, then anything that logically follows from these axioms will also be true, but there's no way that both P and $\neg$ P can be true, for any statement P. On the other hand, it's quite possible for a system to be consistent but not truthful, just as an accomplished liar might be able to present us with an internally consistent story even if it's not true. So truthfulness is a stronger condition than its consistency.

In the story, we first outlined a version of Gödel's theorem saying that if PA is truthful then it is incomplete, and then noted rather briefly that the theorem can be improved so as to require only the assumption that PA is consistent. (The proof of this improved version is given in the endnotes[12].) This approximately mirrors the way things were actually discovered: Gödel originally stated his theorem using a technical condition called *ω-consistency*, which is intermediate in strength between truthfulness and consistency, and the American logician Barkley Rosser later showed how this condition could be replaced by plain consistency. So what we've been calling 'Gödel's first incompleteness theorem' should strictly speaking be called the 'Gödel-Rosser incompleteness theorem'.

There's also another reason for preferring the 'consistency' version of Gödel's theorem to the 'truthfulness' version. This is because the very concept of truthfulness, as we've noted, presupposes the philosophical view that truth for Arithmetic is a meaningful notion, which is not something that would be accepted by the Hilbert finitist school. The concept of consistency, on the

other hand, is meaningful even from a finitist perspective, because the condition '**T** is consistent' can be cast as a *Predictive* statement:

$$\forall x \quad \text{'x is } \textit{not} \text{ a code for a proof of } 0 = 1 \text{ from } \mathbf{T}\text{'}$$

We're here exploiting the idea that an entire proof tree can be coded up in some way by a big Whole number x. And as you might expect, whether a given number x codes a valid proof tree with conclusion 0 = 1 and axioms drawn from **T** is a routinely Checkable property of x. Using this idea, it's possible to state Gödel's theorem in a way that avoids reference to the (for some people philosophically problematic) notion of truth.

Proving the first incompleteness theorem

So much for what Gödel's first incompleteness theorem *says*. Let's now look again at how the theorem is proved.

In the story we gave a high-level outline of a proof of (the 'truthfulness' version of) the theorem in the case of PA. Whilst we covered all the main steps, we suppressed a lot of bureaucratic detail concerning the construction of the Proof Checker and the coding systems involved. We also relied heavily on ideas from Chapter 4, where again a lot of low-level detail was omitted, for instance as regards how statements about tube machine computations can be expressed in Arithmetic. If you have the time and perseverance to work through all the exercises in the supplements on the book's website (in particular, *A Proof Checker for Peano Arithmetic* and *Talking about tube machines in glyph language*), then these will collectively fill in all gaps, and you'll be rewarded with a complete proof of Gödel's theorem. But I hope that the outline given in the story was already enough to be broadly convincing as it stands.

Actually, the route we've taken to the first incompleteness theorem is a bit different from the one Gödel originally took. We've shown how the theorem follows fairly easily from the undecidability of the halting problem, which we covered in Chapter 4. With the advantage of hindsight, this offers a fairly smooth path to Gödel's theorem, but the ideas of Church and Turing weren't around in 1930 when Gödel first proved his result. Instead, Gödel gave a more direct construction of an unresolvable statement, not requiring a detour via tube machines or anything like them.[13] In essence, the two proofs are very similar: in order to construct a glyph statement that states its own unprovability, Gödel used a variant of the same fiendish trick which we used in Chapter 4 to construct a hypothetical tube machine that tested its own halting behaviour. (Or rather, Gödel's version of this trick inspired Turing to use a similar one for the halting problem.)

Gödel's original approach is the one that has been followed in most other existing popular treatments. However, my own experience suggests that many people find the 'Turing route' rather easier to grasp, and it also has the advantage of showing how the various parts of our story all fit together.

One final point, which applies equally to Gödel's proof and ours. It turns out that the entire proof of the first incompleteness theorem can *itself* be conducted entirely within a system such as PA, once we've set up a suitable way to translate statements about proof systems into statements about Whole numbers. As we saw in the story, this is something we exploit in the proof of the *second* incompleteness theorem. But it's also worth mentioning because there's another idea in circulation that Gödel's theorem says something like "a formal system can't reason about itself" — or more specifically, that the proof of Gödel's theorem (for PA, let's say) is something we can't 'understand' without going beyond the limits of PA. But that is simply untrue: the proof of the incompleteness of PA can itself be done in PA. There may, indeed, be other things which we see from outside PA but not from within it, but we'll come on to that in the next chapter.

Examples of unresolvable statements

No sooner do we learn that there are statements that are unresolvable in (for instance) PA, that it's natural to want to see an example of such a statement. Indeed, I remember that when my school maths teacher first told me (simplifying things a bit) that there were mathematical statements that could be neither proved nor disproved, my immediate reaction was "Really?! Such as what?" I don't recall Mr. Orton's exact reply, but it was along the lines of "They're quite complicated."

The particular examples of unresolvable statements that arise directly from proofs of the first incompleteness theorem are indeed highly convoluted and artificial, inhabiting remote and obscure regions of thought that would seem to have little relevance to ordinary mathematics. Gödel's original proof gave rise to a self-referential statement which in effect said 'I am not provable in the system Principia Mathematica' (this was of course carefully couched as a Whole number statement). Our proof via the Turing route, as we saw, indirectly gave rise to another self-referential statement — the Serpent — which effectively said 'I am denied by the Proof Seeker'.

We should pause here to say a bit more about how this 'Serpent' was constructed, as this was something we glossed over in the story. Broadly, the idea is this: *if* our Proof Seeker could resolve all statements of Arithmetic, then in particular, it could resolve all statements about whether computations halted, and so we'd have a Halting Tester. But Turing showed that there is no such thing — but what's more, his proof showed how, for any machine *put*

forward as a Halting Tester, one can construct a *specific example* showing that it isn't. So it's simply a case of running Turing's construction on the hypothetical Halting Tester that would arise from our Proof Seeker, in order to obtain a *specific example* of a halting statement that the Seeker won't be able to resolve.

If you're wanting to follow the details, I've included them in an endnote, which you can read with the help of the tube machine diagram on page Error: Reference source not found.[14] But if not, don't worry: the main point I'm wanting making here is simply that such examples are complex, contrived and quite hard to understand.

There is one important technical sense, however, in which even these very convoluted examples can be classed as 'simple'. Gödel's statement, 'I am not provable', is easily seen to be an example of a Predictive statement. And although our Serpent, 'I am denied by the Proof Seeker', is not itself a Predictive statement, it's easy to see that its negation (¬ Serpent) is provably equivalent to such a statement. This highlights a significant point: the proofs of Gödel's theorem actually show that there are even *Predictive* statements — ones that can't be dismissed as meaningless even by the Hilbert school — that are not resolvable in the systems in question.

We'll see below that there's another class of unresolvable statements whose content is much easier to grasp, namely statements to the effect that certain formal systems are *consistent*. (We've seen already that consistency statements are also examples of Predictive statements.) Even so, these are still highly specialized 'metamathematical' statements about formal languages, proofs and the like, and not likely to be of much interest in themselves, except to that small and suspect tribe of logicians who make it their business to poke around at the very limits of what's provable. So you might very well ask: are there any natural and *non-metamathematical* statements that are expressible in Arithmetic but not resolvable in (lsay) PA? In other words, do the theoretical limitations of PA ever matter for the purposes of ordinary mathematics?

It can be persuasively argued that Gödel's discoveries have had essentially no impact at all on the day-to-day work of (for example) number theorists, who have carried on just as before and haven't come up against any fundamental obstructions to provability in practice.[15] But at the same time, it's interesting to seee just how close we can come to finding simple, intelligible statements of 'ordinary mathematics' that aren't resolvable in PA.

Significant progress on this question had to wait until 1976, when Jeff Paris and Leo Harrington came up with a first plausible candidate for such a statement. Since then, a number of other examples have been discovered, including a few that do undoubtedly have a bearing on deep and important areas of mathematics.[16] But by far the most entertaining example to date was one found by Kirby and Paris in 1982, and this is the one I'd like to describe

here. As you'll see, it still has a rather artificial flavour, but it's strikingly different from the 'metamathematical' examples we've seen so far.

Hercules and the hydra

The example is inspired by an episode from Greek and Roman mythology: the battle of Hercules against the Hydra of Lerna. The Hydra was a serpentine water monster with many heads (the number varies from six to fifty, according to the source) which Hercules had to kill as the second of his Twelve Labours. In some versions of the myth, the Hydra also had the ability to re-grow heads: each time a head was cut off, two (or in some sources, three) new ones would sprout in its place.

The problem we shall consider is an extraordinary elaboration of this idea. For our purposes, a *hydra* will be a many-headed monster whose 'necks' are arranged in a tree-like structure branching from a single root. We'll refer to the branching points (including the root itself) as *nodes*, and to the individual sections of neck (whether between two nodes, or between a node and a head) as *segments*. In any hydra, there are only finitely many heads, nodes and segments altogether. Each node, other than the root, has just a single segment immediately 'below', and each node (including the root) has one or more segments immediately 'above'. (We're using 'below' and 'above' here with the evident meanings of towards and away from the root, in the sense of the tree structure.) Note that we permit nodes with just a single segment above.

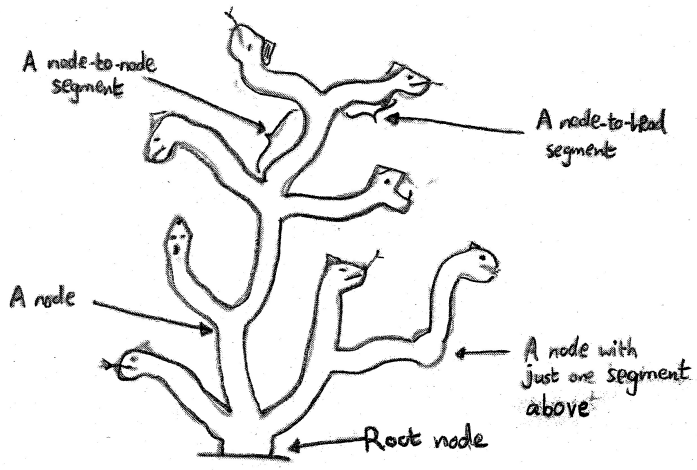


Figure 1: An example of a hydra

In the course of the battle, Hercules tries to decapitate the hydra, one head at a time, in order to reduce it to a single stump (that is, just the root). His problem is that as he does so, the hydra grows new heads in a most alarming fashion, so that it's far from obvious whether Hercules can ever win.

We can think of the battle as proceeding in a series of *rounds*, numbered 1, 2, 3 and so on. On each round, Hercules chops off just a single head — any one he likes. That is, he severs a neck segment connecting a head to some node which I shall call the *severed node*. Actually, we'll suppose Hercules makes his cut cleanly right at the base of this segment, just where it meets the severed node, so that nothing remains of this segment thereafter. As part of the same round, the hydra then 'responds' by growing new heads according to the following rules.

Firstly, if the severed node was the root node, then nothing further happens in this round. What's more, if there are now no remaining neck segments at all, then Hercules has won the entire battle.

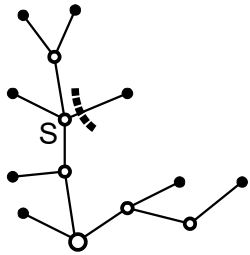
Secondly, if the severed node isn't the root node, and there are now no segments above it, then this node turns into a head. (This isn't mathematically significant — we just add this rule to ensure that every upward path through the hydra terminates in a head.)

But thirdly, if the severed node is not the root node, then (whether or not the second rule applied) the following scary thing happens. Let's refer to the segment immediately below the severed node as the *critical segment*, and to the critical segment plus everything above it — everything from the severed node upwards — as the *critical portion* of the hydra. We'll also refer to the node at the bottom end of the critical segment as the *growth point*, so that the whole critical portion (and perhaps other stuff too) sprouts from the growth point. Now for the scary part. The hydra sprouts several new copies of *the entire critical portion* from the growth point, in addition to whatever was there before. How many new copies exactly? In the Kirby-Paris version, if we were in Round n of the battle, then n copies would be added. However, any decent myth ought to grow a bit in the re-telling, so let's here say that in Round n, the number of new copies of the critical portion will be 2^n.

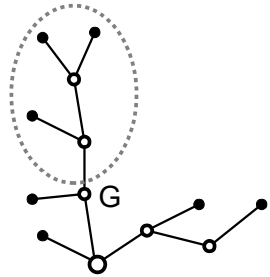
In the first round, then Hercules gets away lightly: only two new copies of the critical portion will be added. (A possible 'Round 1' is shown in Figure 2, starting with the hydra we depicted in Figure 1.) But in the second round, four new copies of the relevant portion will appear; in the third round, eight; and so on. Once these copies have been added, the round is concluded, and the next round begins with Hercules selecting another head to lop off.

So Hercules is faced with an appalling proliferation of heads that appears to get worse and worse as the battle proceeds. Indeed, if you play around with the above rules for a bit, you might rapidly come to the conclusion that, except in rather trivial cases, there's no way that Hercules will *ever* be able to win — he will simply be overwhelmed by the multiplication of heads *ad*

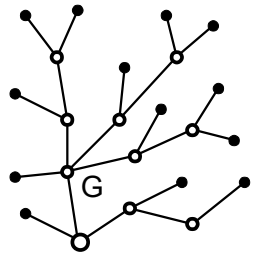
infinitum and will never get a shot at his other ten Labours. In short, the whole contest seems heavily weighted in the hydra's favour.



(a) The previous hydra shown abstractly. The dotted line shows Hercules's first cut, and the severed node is marked 'S'.



(b) One head has been cut off. The growth point is marked 'G', and the critical portion ringed.



(c) Two new copies of the critical portion have now sprouted from the growth point.

Figure 2: A possible 'Round 1' in the hydra battle.
In this round, the number of heads increases from 8 to 13.

But can we *prove* a statement to this effect? Can we prove that Hercules can't win in general? In particular, can we prove such a statement within PA? Of course, for this to be possible, our statement will need to be expressible in Arithmetic. So let's see how we might formulate such a statement, settling for a very modest version in the hope of making our task as easy as possible.

First of all, although our impression might be that 'most' hydras can't possibly be defeated by Hercules, we'll content ourselves with trying to show that there's at least *one* example of a hydra against which Hercules can't win. Secondly, in the interests of presenting the problem in a form that can be easily expressed in Arithmetic, let's suppose that the hydra doesn't even need to stay alive for ever in order to win: at the start of the battle, Hercules names some (preferably large) Whole number *r*, and as long as the hydra can hold out

for *r* rounds, the hydra wins and gets to eat Hercules. Thirdly, we can tilt the scales yet further in the hydra's favour by changing the rules so that, in each round, it's the *hydra* who gets to choose which head Hercules cuts off.

We're now in a position to formulate our statement in Arithmetic. Now that we're familiar with the idea that tube machines, glyph formulae, proof trees and many other 'finite' structures can be represented by Whole numbers using suitable coding tricks, it won't surprise you to learn that hydras, and indeed entire finite battle sequences, can likewise be coded as Whole numbers (you might enjoy working out a way to do this). With this in place, we can now formulate a statement of Arithmetic along the following lines:

$\exists h \ \forall r \ \exists b$ 'b is a battle sequence of r rounds starting with the hydra h'

This says that there's some particular hydra *h* such that, for any number of rounds *r* named by Hercules, the hydra can survive for at least *r* rounds. Note that the hydra in effect gets to determine the whole course of the battle, since it chooses which head to sacrifice at each stage.

Since we've weighted things so much in the hydra's favour, you might think that the statement above was a very mild one, and certainly something we'd expect to be able to prove in PA. Remarkably, however, it turns out that this isn't the case: in fact, it was shown by Kirby and Paris that the above statement is *unresolvable* in PA (assuming, as always, that PA is consistent).

Although this is a somewhat artificial and recreational example, it does show that there are examples of unresolvable statements that don't rely, as the original examples do, on ideas of self-reference or on formal languages, systems of proof, and other metamathematical concepts.

We shall leave Hercules slogging it out against the hydra at this point, but will return to them in the next chapter to see if we can learn anything more about the outcome of their struggle. There may be further surprises in store!

The second incompleteness theorem

Let's turn now to the second of Gödel's incompleteness theorems. Here's how we stated it in the story:

Let **T** *be Peano Arithmetic, or*
any extension of it with a computably defined set of axioms.
If **T** *is consistent, then*
T *cannot prove that* **T** *is consistent.*

As with the first incompleteness theorem, it's possible to generalize the theorem a bit so that it covers axiom systems significantly weaker than PA.

However, in this case it's rather more fiddly to do so, so I shall stick with the statement as it appears above.

Since we've already discussed most of the concepts involved here, this theorem now pretty much speaks for itself. The only point to note concerns the very last line, which you should read as a shorthand for '**T** cannot prove some particular glyph statement which asserts that **T** is consistent.' A few pages back, we mentioned that consistency can be expressed by a statement of Arithmetic of the form:

$$\forall x \quad \text{'x is } \textit{not} \text{ a code for a proof of } 0 = 1 \text{ from } \mathbf{T}\text{'},$$

and under any reasonable formalization of 'being a code for a proof of 0 = 1 from **T**', this will suffice for our present purposes.[17] The rather mind-twisting proof of the second incompleteness theorem for PA that we described in the story then goes through in this more general setting.

The world of consistency statements is certainly a counterintuitive and topsy-turvy one. For instance, the second incompleteness theorem stated above can be rephrased as saying that if **T** can prove that **T** is consistent, then **T** is actually *inconsistent*. But even more bizarrely, perhaps, a set of axioms **T** can declare itself to be inconsistent and yet actually be consistent![18] These apparent paradoxes show just how slippery the terrain is, and how meticulously careful we need to be in our reasoning.

Actually, there's something rather weird about the whole idea of consistency statements, and we can see this even without reference to Gödel's theorems. Let's suppose, just for a moment, that you had genuine doubts about whether PA was consistent. Even if the consistency of PA *could* be proved in PA itself, how would that help you? After all, if PA were *inconsistent*, it would be able to prove anything at all — including the fact that PA is consistent! So in this sense, a proof in PA that PA was consistent wouldn't 'prove' anything, any more than than hearing someone say "I'm telling the truth" convinces you that they're telling the truth. Of course, this point isn't special to PA — the same would apply to any other reasonable system **T** that we're interested in.

So why is the second incompleteness theorem so important? Why is it such a big deal that PA *can't* prove the consistency of PA, if such a 'proof' wouldn't be worth the paper it was written on in any case? To understand the true significance of the second incompleteness theorem, let's return to the story of Hilbert's programme to establish a secure foundation for mathematics, which we began in Chapter 5.

Hilbert's programme scuppered

In general terms, as we saw in Chapter 5, Hilbert's intention was to start from some very restricted system for what he considered as 'real' mathematics (call it ***S***), and to use this to 'justify' the use of some bigger system of 'ideal' mathematics (call it ***T***). We can think of ***S*** here as some system whose principles of reasoning are *so* hard to doubt that they can be taken as 'certain', and of ***T*** as some larger, more convenient system that *might* be considered as doubtful from some philosophical perspectives. In the scenario that Hilbert had in mind, ***S*** was the system we've called *Primitive Recursive Arithmetic* (PRA), whilst ***T*** might be something along the lines of Peano Arithmetic, for example.

Just a quick digression to explain why I've suddenly switched fonts from **T** to ***T*** (although you may let these typographical nuances wash over you if you prefer). Remember that a formal system broadly has three ingredients: a choice of symbolic language (Chapter 3), a set of deduction rules (Chapter 5), and a set of axioms (this chapter). We've so far been using **T** to stand for a set of axioms; but I'd now like to use ***S*** and ***T*** to stand for entire formal systems, comprising all three ingredients. For convenience, we'll assume that ***T*** consists of a set of axioms (which I'll still call **T**) to be used in conjunction with our *standard* language of Arithmetic and Rules of Thought. The system ***S***, however, needn't be of this kind. In particular, we want to allow ***S*** to be a weak system like PRA, which works with a severely cut-down version of our glyph language. When we say that ***T*** is 'bigger' than ***S***, we will just mean that every statement provable in ***S*** is provable in ***T***.

In what sense might ***S*** be used to 'justify' ***T***? Hilbert's observation was that something important would be achieved if we could use ***S*** to prove the *consistency* of ***T***. This is because of a little piece of the jigsaw which we haven't discussed yet: the fact that *consistency is equivalent to Predictive truthfulness*. More precisely:

Let **T** *be any set of axioms such that*
every true Checkable statement is provable from **T**.
Then **T** *is consistent if, and only if,*
every Predictive statement provable from **T** *is true.*

I shall refer to this as 'Hilbert's little theorem' (not standard terminology!). The proof isn't hard and is included in the endnotes;[19] it's worth making the effort to understand it if you have the time and energy to do so.

The point of this little theorem if ***S*** could show that ***T*** was consistent, then anyone who trusted ***S*** could be satisfied that ***T*** was acceptable as a tool for proving Predictive statements.

It's important to grasp just how wonderful this would be if it worked. The Predictive statements, after all, were the only ones that Hilbert was considering as truly meaningful, so such a result would reassure us that we could trust all *meaningful* statements that we could prove in ***T***, however much 'meaningless' stuff was involved in the proofs. And as we saw earlier, the statement '***T*** is consistent' can itself be framed as a Predictive statement, so it's not at all unreasonable to hope that it might indeed be provable in ***S***, if ***S*** was PRA. What's more, a proof in ***S*** that ***T*** was consistent would not be a merely circular piece of self-justification: rather, it would offer genuine reassurance to someone who trusted ***S*** but was initially unsure about ***T***.

Now enter Gödel's second incompleteness theorem. Hilbert is pinning his hopes on using ***S*** to prove '***T*** is consistent', where ***T*** is some larger system than ***S***. But Gödel now tells us that, in all cases of interest, '***T*** is consistent' is not provable even in ***T*** — *let alone in* ***S***! And this single inconvenient fact decisively pulls the plug on Hilbert's programme as he originally conceived it.

As we've seen, the second incompleteness theorem is very general: it establishes that *no* reasonable formal system for mathematics can be proved consistent in any weaker system, or even in itself. And consistency is, after all, the bare minimum that we'd want to know about a system in order to be assured that it was 'safe to use'. So Gödel's theorem spells doom not just for Hilbert's specific project, but for a whole class of similar attempts to provide a secure foundation for mathematics by bootstrapping up from weaker to stronger systems. Just how far-reaching the implications of this are will emerge more clearly over the next few chapters.

The end of certainty?

This is a good time to comment on another popular interpretation of the incompleteness theorems which is slippery at best. In its mild form, this interpretation says something like: 'Gödel's theorems show that we can never be completely certain that the basic principles of mathematics are consistent.' Or in its bolder form: 'The idea of mathematical certainty was demolished by Gödel's theorems.' Although it's easy to see how such ideas might be *suggested* by the second incompleteness theorem, a more careful consideration shows that there's something a bit misdirected about this way of thinking.

The line of thinking that attempts to draw such conclusions from the second incompleteness theorem appears to suppose that the *only* way to attain 'certainty' about some system ***T*** was to prove its consistency (at least) within some 'smaller' system ***S***. But it's easy to see that if we thought *that*, then we'd be faced with an infinite regress anyway, quite irrespective of Gödel's theorems. This is because for such a consistency proof to carry any

conviction, we'd need to be sure that ***S*** was at least consistent. And if the only way to be sure of that was to prove it in some yet smaller system ***R***, then we'd need to be assured that ***R*** was consistent. And so on *ad nauseam*.

This, in fact, boils down to nothing more than the completely simple point that's familiar to every child who discovers that a good way to annoy adults is to keep on asking "Why?" or "How do you know?" until they're forced to shut up. Obviously, a 'proof' of any kind — in mathematics or elsewhere — can't get off the ground unless we're willing to accept something *without* proof. That's something we've known at least since Aristotle's time; it didn't take Gödel to tell us that!

I think that before about 1890, it wouldn't have occurred to any mathematician that their belief in the certainty of mathematics *needed* to be backed up by anything like a consistency proof. In the early 20th century, to be sure, the emergence of various 'paradoxes' in set theory caused mathematicians to lose their nerve for a time, and this led Hilbert and others to seek the reassurance that a consistency proof might offer. (Even Hilbert's project, however, took for granted that Primitive Recursive Arithmetic was reliable beyond question and didn't in itself require justification.) By the mid-20th century, mathematicians had arrived at a much clearer way of understanding set theory which resolved the paradoxes and restored confidence in the mathematical edifice as a whole (we shall come to this in Chapter 9). But this was achieved without the help of anything like a consistency proof. So from this perspective, the whole idea that 'mathematical certainty' (in the traditional sense) depended on being able to ground the mathematical edifice on some more primitive foundation appears historically as a somewhat curious suggestion that enjoyed only a brief heyday. Mathematics, as it turns out, simply doesn't have the sort of 'ultimate foundation' that the likes of Russell and Hilbert were looking for, but it is none the less solid and robust for that. It is in this sense that one might liken mathematics to a 'castle in the air'.

Ultimately, insofar as 'certainty' can be taken to refer to a subjective psychological state, it's hard to say anything very definitive about what is and isn't certain. If someone says that reflecting on Gödel's theorems makes *them* feel less certain about mathematics than they were before, they may be right: indeed, these theorems temporarily did that for me when I first encountered them. But no such loss of confidence is forced on us by the theorems themselves. There's nothing in these theorems, for example, to challenge the simple, honest belief that the axioms of PA say obvious and unquestionably true things about the Whole numbers, and that we can simply 'see' this directly. But nor do the theorems offer any particular support for this view. In fact, Gödel's theorems turn out of be perfectly compatible with a great many different philosophical positions.

Indeed, one might even say that far from answering any of the really big questions, Gödel's theorems raise even more questions, in that they open the door to a far wider range of philosophical options than we might previously have imagined. This will become clearer as we continue our journey.

Relative consistency

It might seem at this point that the whole topic of consistency proofs is a rather barren one: a proof in ***S*** that ***T*** is consistent will be impossible if ***S*** is smaller than or the same as ***T***, and worthless if ***S*** is larger than ***T***. Actually, that's not quite the whole story: there are interesting cases where ***S*** is neither larger nor smaller than ***T***, but just *different*, and in such cases a consistency proof can indeed tell us something informative.[20]

But there's also another direction we can go in. Let's suppose ***S*** is smaller than ***T***. Although ***S*** won't be able to prove that ***T*** is consistent, we can ask whether ***S*** might be able to prove the next best thing, namely:

If ***S*** *is consistent, then* ***T*** *is consistent.*

Or to frame this in a slightly more robust way, we can ask whether ***S*** extended with the additional axiom '***S*** is consistent' can prove '***T*** is consistent'. If it could, then we'd at least know that ***T*** was no more likely to be inconsistent than ***S*** was. So if we were comfortable with ***S*** and willing to accept that it was truthful (hence consistent), we could bootstrap ourselves up to the belief that ***T*** was consistent — which, as we noted earlier, would at least tell us that ***T*** was truthful as far as Predictive statements were concerned.

There are quite a few situations in which this 'modified Hilbert's programme' works well, and can indeed be used to justify some non-trivial bootstrapping. We'll meet some interesting examples of this later in the book. Unfortunately, though, in the typical cases that Hilbert originally had his eye on, it turns out that even this more modest alternative to his original idea won't work.

The idea is this. Suppose we're in a situation where ***T*** is not only larger than ***S***, but is powerful enough to prove '***S*** is consistent'. You might like to think here of ***T*** as representing some higher vantage-point from which we can 'look down' on the poor benighted souls who are confined to working in ***S***, and can see things about ***S*** that they themselves can't see. This is a very common situation for the sorts of systems we're interested in, and in particular it holds for our paradigm case: PA can quite easily prove the consistency of Hilbert's PRA.

Suppose, then, that ***T*** can prove '***S*** is consistent', where ***T*** is PA of some computably generated extension of it. In this case, it's quite easy to see that ***S***,

or even ***T***, can't get us from '***S*** is consistent' to '***T*** is consistent'. For if it could, then by combining these two proofs, we'd get a proof in ***T*** that ***T*** was consistent. Which, by the second incompleteness theorem, is impossible!

In other words, if the gap between ***S*** and ***T*** is wide enough that ***T*** can 'see' that ***S*** is consistent, then we can't hope for even a *relative* consistency result to bootstrap from ***S*** to ***T***. In such a case, the gap is of a fundamental character and cannot be bridged. Logicians express this by saying that ***T*** has greater *consistency strength* than ***S***.

Let's reflect a little more on what this means in our paradigm case, where ***S*** is PRA and ***T*** is PA.

Rifts in the landscape

We noted earlier in this chapter that there's a philosophical difference between Hilbert's 'finitistic' view of mathematics, which underpins PRA, and the 'classical' view of most mathematicians, which underpins PA. This difference, as we saw, can be encapsulated by what we called the surveyability of $\mathbb{N}$. Whilst for most mathematicians this principle might be entirely non-controversial, we've seen that there's also a perspective from which it might be seen as doubtful — suggesting that it would be desirable, if it were possible, to give some justification for the use of PA that didn't rely on this principle.

What we've now seen is that this can't be done. More specifically, that there's *no* way to bootstrap ourselves up from trust in PRA up to trust in PA. There is, so to speak, an unbridgeable gulf between these systems — a deep rift cutting right across the mathematical landscape.

One might wonder whether one could get round the problem by replacing PRA with some more powerful system that can still be justified on the basis of a 'finitist' philosophy. Indeed, Gödel himself, at the end of his 1931 paper, floated the suggestion that something along these lines might be possible. However, he soon changed his mind, and nowadays it's generally agreed that *no* system that could plausibly deserve the label 'finitist' principles would overcome the basic problem: there would always be systems acceptable to the classical mathematician whose (relative) consistency would remain out of reach to the finitist. In other words, the mathematical consequences of the surveyability of $\mathbb{N}$ have to be accepted via a 'leap of faith', or not at all.

It will turn out that this isn't the only rift to cut through the mathematical terrain in this way: indeed, this particular rift is simply the first example of a recurring phenomenon. At the moment, we're supposing that Anna and Emily accept the surveyability of $\mathbb{N}$ without question, and are blissfully unaware of the 'lower world' of Hilbert's finitism and of the gulf between that and their own world. However, it won't be long before they encounter another such

gap, separating them from a still 'higher' realm which they're unable to reach on the basis of their current assumptions.

This picture requires a bit of qualification. As we shall learn soon, there are gaps and there are Gaps. In other words, there are 'mild' situations where some system $\boldsymbol{T}$ *formally* has greater consistency strength than a smaller system $\boldsymbol{S}$, but *philosophically* the validity of $\boldsymbol{T}$ is really no harder to accept than that of $\boldsymbol{S}$. But there are other situations in which the Gap is of a much more serious nature: in these cases, some definite new principle or 'belief' is needed to make the leap from $\boldsymbol{S}$ to $\boldsymbol{T}$. Naturally, it's the latter sort — the Gaps — that are the more interesting. In the course of the next few chapters, we shall encounter several more such Gaps and discover what kinds of philosophical principles are needed to cross them.

The overall picture, then, is of a mathematical landscape crossed by a number of rifts which cannot be bridged and which there is simply no way round: in each case, some new principle needs to be accepted on trust in order to reach the other side. This understanding of the nature of mathematics, and of the inescapable existence of these gaps, is perhaps the main thing to have been revealed by the light of Gödel's incompleteness theorems.

There are also other interesting repercussions of Gödel's theorems, and we'll point out some of these as we go on. But to enable us to continue the journey we have just outlined, the time has come to introduce a new idea — the concept of *reflection* — and in connection with this, to consider closely what may be the most famous of all purported applications of Gödel's theorems outside mathematics itself: the claim that these theorems reveal something important about the nature of the human mind.

[1] There are several minor differences of detail between Peano's version of the axioms and ours (for instance, Peano chose to consider the Whole numbers starting from 1 rather than 0). But there is also a more serious difference: in technical language, we've adopted a *first order* induction scheme, whereas Peano originally used a *second order* formulation of the induction axiom (this will all be explained in Chapter 8). In spite of this, it's the first order version of the system that is most commonly known nowadays as 'Peano Arithmetic'.

[2] In Wittgenstein (1953), §352. Of course, given how many digits of π have now been computed, one would nowadays need a longer sequence of '7's for this example to retain its point.

[3] A textbook example is the proof of the well-known formula for the *n*th *triangular number*:

$$1 + 2 + \ldots + n = n \times (n + 1) / 2.$$

This is easily checked when $n = 0$ (or you can start at $n = 1$ if you like). And assuming it's true for n, it's also true for $n + 1$, because:

$$\begin{aligned} 1 + 2 + \ldots + n + (n + 1) &= n \times (n + 1) / 2 + (n + 1) \\ &= (n + 1) \times (n + 2) / 2 \end{aligned}$$

with the help of a bit of school algebra. So by induction on n, we can conclude that the formula is true for every Whole number n.

[4] Notice that we're here taking the operation '+1' to be more fundamental than the *general* operation of addition. This is admittedly courting confusion, and in most logic texts, the point is clarified by using a different notation for the operation of adding 1: the Whole number after x is written as Sx ('the successor of x'). But I'm trying to get away with as little specialized notation as possible.

[5] The debate concerns not only what kinds of reasoning the Hilbert school intended to allow (and some argue that they intended different things at different times), but also which of these are actually justifiable on the philosophical basis they claimed to be adopting. Tait (2005) argues strongly for the view that PRA captures precisely the reasoning that is admissible from a finitist perspective, on any coherent interpretation of the latter. Against this, Kreisel claimed that more than PRA is finitistically justifiable, and indeed proposed a hierarchy of

'finitistic' systems that in some sense creep up on PA in their strength. (See G. Kreisel, *Ordinal logics and the characterization of informal concepts of proof*, in J.A. Todd (ed.), *Proceedings of the International Congress of Mathematicians, 1958*, Cambridge University Press (1960).) I think Tait is right, though I'll confess that I find the whole business maddeningly hard to think about. The difficulty lies in relentlessly purging all tempting non-finitistic inferences from one's mind, and in squeezing oneself into the almost unbearably constricted finitist mindset.

[6] I won't give a full definition of PRA here, but here's a bit more information. First, all of the axioms we've given for PA, with the exception of those given by the induction scheme are easily seen to be Predictive statements. So we can simply strip off the '∀' prefixes from these axioms to obtain Checkable formulae, and let these to stand as axioms of PRA. For example:

$$\odot \ \neg\, x + 1 = 0$$

Secondly, we add a facility for introducing further 'primitive recursive' functions such as ^ (exponentiation) and ! (factorial). I won't explain this in generality here, but in the case of ^ and !, the effect is that the equations given in endnote [?] will be admitted as axioms that define these operations. This should give at least a flavour of the general mechanism.

Thirdly, in view of the fact that we're using $\neg\, x + 1 = 0$ (for example) to do duty for $\forall x \ \neg\, x + 1 = 0$, we'll need a corresponding rule that does duty for the ∀ Elim rule:

$$\frac{\mathbf{A} \odot P \qquad v \qquad T}{\mathbf{A} \odot P\,[v \rightarrow T]}$$

Finally, and most interestingly, we note that the induction axioms of PA don't have the form of Predictive statements, so we'll need another way of getting induction into our system. For this purpose, we introduce the following special *induction rule*:

$$\frac{\mathbf{A} \odot P[z \rightarrow 0] \qquad \mathbf{A} \odot P[z \rightarrow x] \Rightarrow P[z \rightarrow x+1]}{\mathbf{A} \odot P} \quad z,x \text{ not free in } \mathbf{A}$$

Here P and all the formulae of **A** must of course be Checkable, in line with our blanket restriction in PRA to Checkable formulae.

[7] For an authoritative discussion, including a proposed outline for a proof of Fermat's Last Theorem within PA, see A. Macintyre, *The impact of Gödel's incompleteness theorems on mathematics*, in Baaz et al (2011). See also C. McLarty, *What does it take to prove Fermat's Last Theorem?*, in *Bulletin of Symbolic Logic* 16(1), 359-377 (2010).

[8] This is a variation on a point made by Georg Kreisel in 1974: '...even today we do not have any theorem in ordinary number-theoretic practices which cannot be proved in PM. And though this empirical fact does not contradict incompleteness (since there are many open problems in ordinary number theory), it may seem to support the following hypothesis...: if a number-theoretic proposition can be proved at all, it can be proved in PM.' (See G. Kreisel, *A notion of mechanistic theory*, in *Synthese* 29, 11-26 (1974).) By 'PM' Kreisel means the system Principia Mathematica of Russell and Whitehead, a stronger system than PA which can naturally accommodate proofs via Complex analysis and the like. But essentially the same point can be made (admittedly with a little less certainty) for PA.

Kreisel's point here was that cumulative evidence based on 'naturally arising' examples can be misleading, and one might need to look in obscure places (as in mathematical logic) for examples that show what's really going on. This is a point we alluded to in Chapter 4: from this perspective, cumulative evidence for the Church-Turing thesis does not seem particularly strong evidence!

[9] The first incompleteness theorem was announced by Gödel at a conference in Königsberg in September 1930. Both incompleteness theorems were presented in his landmark paper, 'On formally undecidable propositions of *Principia Mathematica* and related systems I', submitted in November of that year and published early in 1931. The paper is reproduced in Van Heijenoort (1967) and in Gödel (1986).

[10] As amusing examples of spurious appeals to Gödel's theorems, based apparently on 'some process of free association', Franzén quotes the following: 'By Gödel's incompleteness theorem, all information is innately incomplete and self-referential', and 'By equating existence and consciousness, we can apply Gödel's incompleteness theorem to evolution.'

[11] In Gödel's 1931 paper, the incompleteness theorems were not formally stated in a particularly general form, because an appropriate general concept of a 'computable set of axioms' was not then available. Later, however, Gödel embraced Turing's 1936 discoveries as leading to 'a precise and unquestionably adequate definition of the general notion of formal system', and thus to a broader statement of his incompleteness theorems (see Gödel's 1963 addendum to his 1931 paper, reproduced on page 616 of Van Heijenoort (1967)).

[12] Here, for the sake of (in)completeness, is the argument that shows 'if PA is consistent, then PA is incomplete'. For this, we'll need to use the facts about the 'Serpent' statement established in endnote [?] below (whose proof is independent of this one).

We shall actually show: 'If PA is consistent, the Serpent is unresolvable in PA'. We'll argue by assuming the Serpent *is* resolvable in PA, and heading for the conclusion that PA is inconsistent. We'll write S for the Proof Seeker, and s for the numerical code for the Serpent (the yellow-green number in the story).

Recall that 'resolvable in PA' means 'either provable or disprovable in PA (or perhaps both)' — which implies that the Proof Seeker will eventually be able to discover a proof or disproof. So saying that the Serpent is resolvable in PA implies the following: if the Proof Seeker S is given the number s, the computation will eventually terminate, either with the discovery of a proof of the Serpent or with the discovery of a disproof. Let's consider these two cases separately.

Firstly, suppose this computation terminates with the discovery of a proof of the Serpent in PA — that is, the disc leaves S at the first exit. Since this computation K is just a finite sequence of steps, the statement

S, when fed with s, yields the computation K
which terminates at the first exit

is an example of a Checkable statement, in the terminology of Chapter 5. Since it is also a true statement, it is provable in PA:

⊙ 'S, when fed with s, yields the computation K
which terminates at the first exit'

From this it follows that PA can prove:

⊙ 'S, when fed with s, doesn't yields a computation

which terminates at the second exit'

or in other words

⊙ ¬ 'The statement coded by s is denied by S'

Now, by endnote [?], the sentence in quotes here is provably equivalent to the Serpent itself! So PA can prove:

⊙ ¬ Serpent

But we're assuming, in this first case, that the Proof Seeker has discovered a proof of the Serpent in PA. So PA proves both the Serpent and its negation, and thus PA is inconsistent.

Now for the second case. Suppose that when S is fed with s, the computation finishes with the discovery of a disproof of the Serpent in PA — that is, the disc leaves S at the second exit. Once again, this computation K is just a finite number of steps, and so

S, when fed with s, yields the computation K
which terminates at the second exit

is a true Checkable statement, and hence is provable in PA. So PA can prove:

⊙ 'The statement coded by s is denied by S'

So appealing once again to endnote [?], PA can prove the Serpent. However, in this second case, we are assuming the Proof Seeker has discovered a *disproof* of the Serpent: that is, a proof of ¬ Serpent. So once again, PA is inconsistent.

Thus, we've shown that PA is inconsistent in either of these cases, and that completes the proof.

[13] Gödel's original method worked like this. First, we note that if p is a Whole number that codes a glyph formula P which itself contains a free variable x, then we can get a new formula P[x→p] by replacing all free occurrences of x by the actual number p. This is the glyph statement equivalent of the SELF machine: the idea is that we're 'applying' the formula P to its own code p. Let's now write f(p) to mean the Whole number code for this new formula P[x→p]; note that f(p) can be calculated solely from p (since p is both a code for P and the value

we're replacing x by).

Next — and this is where the hard work comes — we construct a glyph formula G, featuring a variable x, whose meaning is 'The glyph statement coded by the number f(x) is not provable in **T**.' We're now set up to get a 'self-biting' statement as follows: let g be the Whole number code for the formula G, and consider the statement G[x→g], whose code is f(g). This statement, in effect, says 'I am not provable in **T**.' It's now not hard to show that if **T** is truthful (or in Gödel's original version, *ω-consistent*), then this latter statement is unresolvable in **T**.

[14] Here are the details. With reference to the tube machine diagram on page Error: Reference source not found, we need to know the following. The *Halting Statement Builder* takes the numbers in the A and B registers, constructs the glyph statement that says 'Tube machine A halts on inputs B', and puts the numerical code for this glyph statement in the C register. The *SELF Machine*, takes the number *a* given in the A register, and sets up B so as to code a register disc with A = *a* and all other registers zero (see the Chapter 4 story for a more gentle explanation!) We'll refer to the combination of the SELF Machine with the Halting Statement Builder as the SMHSB.

The construction of the 'Serpent' went as follows. Let M be the whole tube machine shown on page Error: Reference source not found, and let m be the numerical code for this machine. By feeding m through the SMHSB, we obtain a certain number s in the C register. By what we've said above, s will be a numerical code for a glyph statement that may be paraphrased as:

'The machine coded by m halts when given the number m'

and the Serpent is this glyph statement itself. It's now not hard to see that the Proof Seeker cannot (correctly) return either Yes or No when fed with s: if it returned Yes, then M would fail to halt when fed with m, and the Serpent would be false; whereas if it returned No, then M would halt when fed with m, and the Serpent would be true.

This gets us as far as establishing that the Serpent is indeed unresolvable by the Proof Seeker, assuming the Proof Seeker is truthful.

But we also want to justify our paraphrase of the Serpent's content as 'The Proof Seeker denies this statement'. More precisely, we can show that the following two statements are provable in PA:

- Serpent ⇒ 'The Proof Seeker denies the statement coded by s'
- 'The Proof Seeker denies the statement coded by s' ⇒ Serpent

Thus, the statement in quotes here, whilst not literally identical to the Serpent, is provably equivalent to it in PA — which is good enough for our purposes.

Let's see how these two implications are provable. We've already noted what the Serpent actually says may be expressed as:

'The machine coded by m halts when given the number m'

Furthermore, there is some finite SMHSB computation (call it c) that produces s from the input m. Now, the statement

'c is an SMHSB computation which yields the output s
from the input m'

is an example of a true Checkable statement, and so it's provable in PA. Using this, it is easy to see that the Serpent itself, i.e. the statement

'The machine coded by m halts when given the number m'

can be proved equivalent (in PA) to the statement

'The Proof Seeker terminates at the second exit
when given the input s'

which in turn can be proved equivalent (in PA) to the statement

'The Proof Seeker denies the statement coded by s'

This establishes the two implications claimed above.

[15] See A. Macintyre's article in Baaz *et al* (2011) for powerful advocacy of this view.

[16] Perhaps the most striking example to date is a finite version of the Robertson-Seymour *graph minor theorem*, a very deep result in graph theory. See H. Friedman, N. Robertson and P. Seymour, *The metamathematics of the graph minor theorem*, in *Logic and Combinatorics*, Contemporary Mathematics Series 65 (1987). Numerous other examples of statements unprovable in PA appear in

Harvey Friedman's recent writings.

[17] One can get into trouble if one chooses to translate '**T** is consistent' into glyph language in a particularly wonky way. But for all the systems we'll be interested in, any of the 'obvious' ways of doing it will be fine, so one shouldn't worry unduly over this subtle point. The situation was analysed in S. Feferman, *Arithmetization of mathematics in a general setting*, Fundamenta Mathematicae 49 (1960).

[18] You might well think that *this* is inconsistent! But actually, an example of this is quite easy to give. Let **T** consist of the axioms of PA augmented with a single new axiom that says 'PA is inconsistent'. I claim that as long as PA is consistent (which we all believe), then **T** is consistent. For if it weren't, then we would in effect have a proof within PA that the assumption 'PA is inconsistent' leads to a contradiction — from which we get a proof within PA that PA is consistent! But this is impossible by the second incompleteness theorem.

So that shows that **T** is consistent (assuming PA is). But it's easy to see that **T** declares itself to be inconsistent: **T** contains the axiom 'PA is inconsistent', from which it's easy to prove '**T** is inconsistent', because **T** is a larger theory than PA.

Of course, a consistent theory that thinks it's inconsistent cannot possibly be a *truthful* theory. Indeed, **T** is manifestly not truthful, because no one believes the statement 'PA is inconsistent' to be true. So our **T** is also interesting as an example of a theory that's consistent but not truthful.

[19] Let's prove Hilbert's little theorem.

One direction is very easy. If **T** were inconsistent, then anything would be provable in **T**, including blatantly false Predictive statements such as $\forall x \;\; x = 0$. So 'every Predictive statement provable from **T** is true' certainly implies '**T** is consistent'.

What about the other direction? For this, let's suppose (hypothetically) that **T** isn't Predictively truthful: that is, that some Predictive statement P that's provable in **T** is actually false. We saw in Chapter 5 that a Predictive statement has the form of a Checkable statement prefaced by some number of '$\forall$' prefixes. For simplicity, we'll here just consider the case of a single prefix, so we can suppose P has the form $\forall x \; Q$ where Q is a Checkable formula involving x. (The general case requires more notation, but isn't really any harder.)

We're supposing, then, that $\forall x\ Q$ is provable from **T**, but is actually *false*. What does this mean? It means that *not all* numbers x satisfy the property Q. So there's some particular number, n say, such that Q[x→n] (that is, Q with the number n substituted for the variable x) is false.

Which in turn means that $\neg\ Q[x{\rightarrow}n]$ is a true statement. And since Q is a Checkable formula, this is also a Checkable statement. But our theorem assumes that **T** is strong enough to prove every true Checkable statement. So $\neg\ Q[x{\rightarrow}n]$ is provable as a theorem from **T**.

But wait a minute! We've already said that $\forall x\ Q$ is provable from **T**. And from this, by a single application of the $\forall$ Elim rule, we can deduce Q[x→n]. So Q[x→n] is also provable from **T**.

So we've now shown that both $\neg\ Q[x{\rightarrow}n]$ and $Q[x{\rightarrow}n]$ are provable from **T**. So **T** is inconsistent!

To summarize, we've just shown that if **T** is not Predictively truthful, then **T** is inconsistent. Or to put it the other way round, if **T** is consistent, then **T** is Predictively truthful. So we've now established both directions of what is claimed by Hilbert's little theorem.

A version of this argument (using the statement of Fermat's last theorem as an illustrative example) was presented by Hilbert in *The Foundations of Mathematics* (1927); see page 472 in Van Heijenoort (1967).

You might wonder in which system the above proof itself can be conducted. (Indeed, if you are learning to think like a logician, then you *should* wonder this!) As I've presented it above, the theorem just exceeds what can be stated in PRA, though arguably not in a very serious way. Specifically, one can explicitly define a finitistically acceptable operation on proofs which does the following: given a proof in PRA that **T** is consistent, and a proof in **T** of some Predictive statement P, it delivers a proof in PRA of P, which the finitist can then understand and believe. The only snag is that the statement I've just made cannot *itself* be seen by the finitist to be true. Strictly speaking, then, after proving P in **T**, the finitist will need to go through the rigmarole of applying this operation to get a proof of P in PRA. *We*, as non-finitists, can see that this will always work, and that the finitist will always be satisfied with the result. But the finitist will actually have to go through this process to be assured of the theorem P.

[20] As a classic example, Gerhard Gentzen in 1934 gave a proof of the consistency of PA conducted in Hilbert's Primitive Recursive Arithmetic extended with a certain principle of *transfinite induction*. In other words, the loss of strength that comes from cutting down to just Predictive statements can be compensated for by adopting a stronger

induction principle than the usual one. Whether you think that this 'transfinite induction' is any less doubtful than anything in PA itself is ultimately a matter of personal judgement — but at least it's *different*, and offers an alternative slant on what it means to trust PA.